

U.S. Futures and Options Package

For Business: Friday, June 09, 2023

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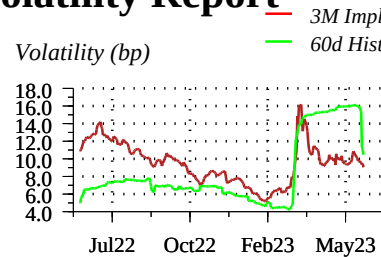
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Short-Dated SOFR Swaption Volatility Report

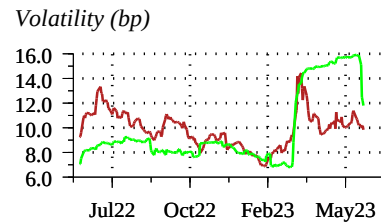
Options on 1yr Swaps

Mat	Implied Vol.			5d	20d	Historical BP Vol.		
	%	bp	Chg			10d	20d	60d
1m	29.89	9.49	-1.09	-0.94		6.71	5.35	10.47
3m	31.37	9.54	-1.02	-1.23		*1dc	1wc	2wc
6m	34.58	9.68	-0.79	-0.97		5.35	4.75	6.16
12m	42.21	9.79	-0.59	-0.84				



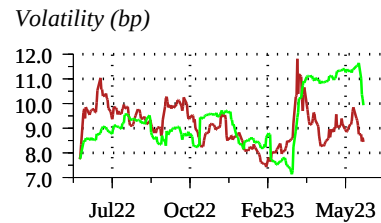
Options on 2yr Swaps

Mat	Implied Vol.			5d	20d	Historical BP Vol.		
	%	bp	Chg			10d	20d	60d
1m	38.47	10.51	-0.77	-0.52		9.94	7.61	11.80
3m	39.17	10.23	-0.79	-1.01		*1dc	1wc	2wc
6m	40.49	9.85	-0.62	-0.83		7.61	6.27	7.65
12m	43.14	9.21	-0.44	-0.77				



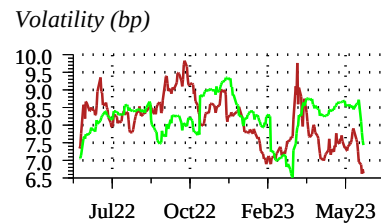
Options on 5yr Swaps

Mat	Implied Vol.			5d	20d	Historical BP Vol.		
	%	bp	Chg			10d	20d	60d
1m	38.88	8.80	-0.79	-0.25		9.67	7.42	9.93
3m	38.89	8.60	-0.69	-0.61		*1dc	1wc	2wc
6m	38.57	8.23	-0.51	-0.44		7.42	6.27	6.59
12m	38.55	7.75	-0.39	-0.34				



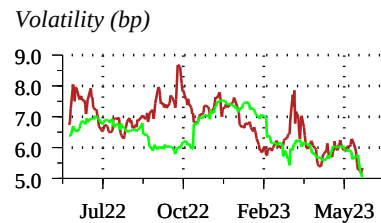
Options on 10yr Swaps

Mat	Implied Vol.			5d	20d	Historical BP Vol.		
	%	bp	Chg			10d	20d	60d
1m	30.75	6.64	-0.59	-0.60		7.84	6.08	7.43
3m	31.05	6.63	-0.60	-0.79		*1dc	1wc	2wc
6m	31.67	6.64	-0.48	-0.52		6.08	5.28	4.66
12m	32.40	6.60	-0.33	-0.29				

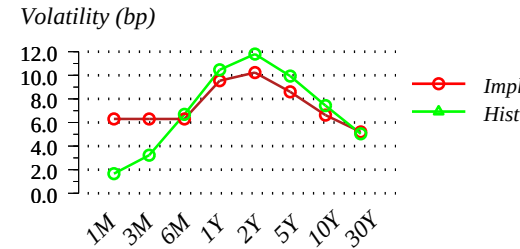


Options on 30yr Swaps

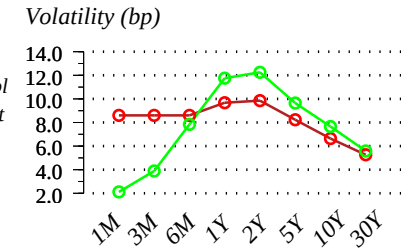
Mat	Implied Vol.			5d	20d	Historical BP Vol.		
	%	bp	Chg			10d	20d	60d
1m	25.66	5.16	-0.04	-0.61		5.01	4.31	5.04
3m	26.08	5.20	-0.18	-0.72		*1dc	1wc	2wc
6m	26.70	5.26	-0.25	-0.52		4.31	4.02	3.18
12m	27.26	5.27	-0.22	-0.40				



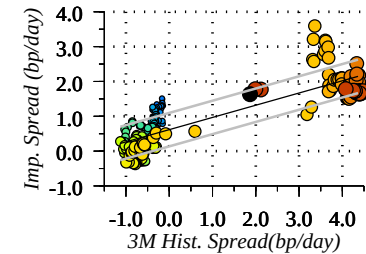
Maturity Structure of 3M Vol.



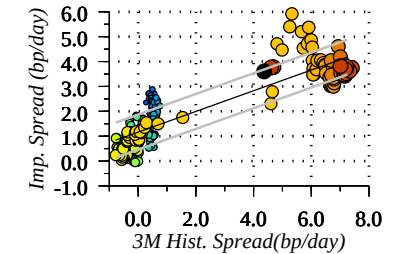
Maturity Structure of 6M Vol.



3Mx2Y - 3Mx5Y



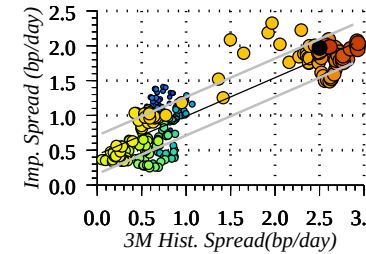
3Mx2Y - 3Mx10Y



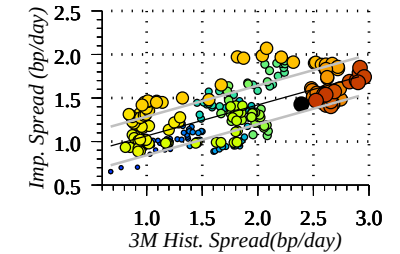
$$Y = 0.3508 X1 + 0.6160 \text{ (adjR2 = 0.6686)}$$

$$Y = 0.4159 X1 + 1.1630 \text{ (adjR2 = 0.7288)}$$

3Mx5Y - 3Mx10Y



3Mx10Y - 3Mx30Y



$$Y = 0.5534 X1 + 0.4351 \text{ (adjR2 = 0.7434)}$$

$$Y = 0.3561 X1 + 0.7087 \text{ (adjR2 = 0.4786)}$$

Mat/Und	Hist Vol.		Implied Vol.	
	hold	curr	hold	curr
3mx1y	3.52	10.47	4.08	9.54
3mx2y	4.47	11.80	4.65	10.23
3mx5y	5.25	9.93	5.11	8.60
3mx10y	5.40	7.43	5.21	6.63
3mx30y	5.08	5.04	4.98	5.20

1. Historical BP Vol. is calculated the 10-day, 20-day, and 60-day standard deviations of daily spot SOFR rate changes on the tenor specified in the table.

2. Regression of difference in 3M implieds vs difference in 3M historicals.

3. *One-month realized BP Vol. over 1-day, 1-week and 2-week change.

Derivatives Strategy

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	Treasury Bond				Treasury Note				5 Year Treasury Note			
	Jul 23	Aug 23	Sep 23	Dec 23	Jul 23	Aug 23	Sep 23	Dec 23	Jul 23	Aug 23	Sep 23	Dec 23
VOLATILITY												
Implied (%)												
Current	11.07	11.12	11.40	11.51	7.24	7.10	7.34	7.41	5.89	5.61	5.69	5.65
Change (1d)	-0.34	-0.21	-0.15	-0.16	-0.05	-0.11	-0.03	-0.04	0.08	0.05	0.02	0.03
20 day High	13.96	13.58	13.64	13.06	9.10	8.64	8.71	8.31	7.17	6.71	6.66	6.23
20 day Low	10.66	10.80	11.12	11.22	7.20	6.98	7.21	7.25	5.76	5.51	5.60	5.57
Business Day	11.24	11.22	11.35	11.52	7.35	7.16	7.31	7.42	5.98	5.66	5.67	5.66
Historical (%)												
10 day	-	-	-	-	8.10	8.10	8.10	8.14	6.13	6.13	6.13	-
20 day	-	-	-	-	6.34	6.34	6.34	6.37	4.74	4.74	4.74	-
50 day	-	-	-	-	7.18	7.18	7.18	7.25	5.63	5.63	5.63	-
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)												
3 Year High	426.01	241.69	175.79	119.48	15.96	11.39	9.93	8.55	13.26	9.22	7.90	6.37
3 Year Average	16.93	20.11	21.95	16.15	5.69	5.80	5.76	4.58	3.84	3.92	3.81	2.89
3 Year Low	3.05	4.88	5.34	3.63	1.11	1.75	2.01	1.37	0.39	0.85	0.99	0.67
% Historic Above Implied	39.00	46.64	49.01	26.75	29.91	39.66	37.29	12.52	18.18	28.99	26.35	5.01
BASIS POINT VOLATILITY												
Implied (bp)												
Current	5.8	5.9	6.0	6.1	7.8	7.6	7.8	7.2	9.6	9.1	9.2	9.1
Change (1d)	-0.2	-0.1	-0.1	-0.1	-0.0	-0.1	-0.0	-0.0	0.1	0.1	0.0	0.0
Change (5d)	-0.2	-0.1	-0.2	-0.0	-0.7	-0.6	-0.4	-0.1	-0.7	-0.7	-0.6	-0.0
Change (20d)	-0.6	-0.6	-0.6	-0.7	-0.7	-0.9	-0.7	-0.3	-0.7	-0.8	-0.8	-
20 day High	7.2	7.1	7.1	6.8	9.7	9.2	9.2	8.0	11.7	10.9	10.7	10.1
20 day Low	5.6	5.7	5.9	5.9	7.7	7.5	7.7	7.0	9.3	9.0	9.1	9.0
Historical (bp)												
10 day	4.9 *	4.9 *	4.9 *	4.9 *	8.5	8.5	8.5	8.5	9.5	9.5	9.5	9.5
20 day	4.3 *	4.3 *	4.3 *	4.3 *	6.7	6.7	6.7	6.7	7.3	7.3	7.3	7.3
50 day	4.9 *	4.9 *	4.9 *	4.9 *	7.5	7.5	7.5	7.5	8.5	8.5	8.5	8.5
Volatility Cones (bp)												
3 Year High	12.5	9.8	9.5	8.6	17.0	12.0	10.4	8.7	20.5	14.3	12.2	10.3
3 Year Average	5.6	5.7	5.7	5.9	5.8	5.8	5.7	5.1	5.6	5.7	5.7	5.7
3 Year Low	1.5	2.5	3.2	3.8	0.9	1.7	2.1	2.8	0.7	1.2	1.6	2.0
% Historic Above Implied	54.8	58.0	53.9	51.4	36.1	41.4	37.4	31.8	15.0	24.5	21.9	22.4
Yield Change Required to... (bp)												
Offset 1 C-day Decay	4.0	4.0	4.2	4.2	5.3	5.3	5.5	5.1	7.2	7.0	7.2	7.2
Offset 1 B-day Decay	4.8	4.9	5.0	5.1	6.4	6.3	6.6	6.2	8.7	8.4	8.7	8.7
Offset 1 Month Decay	-	20.8	22.4	23.0	-	25.6	28.9	27.7	-	34.8	38.1	39.2
Break Even at Expiry	12.3	21.0	29.2	43.3	16.5	27.6	38.8	53.1	22.6	36.7	50.8	74.7
BPV per Contract	\$185.68	\$185.68	\$185.68	\$190.53	\$80.79	\$80.79	\$80.79	\$90.14	\$46.10	\$46.10	\$46.10	\$48.19
Cal/Bus Days to Expiry	15/10	43/29	78/54	169/116	15/10	43/29	78/54	169/116	15/10	43/29	78/54	169/116
Futures Price	127-20	127-20	127-20	127-21	113-235	113-235	113-235	114-075	108-17	108-17	108-17	109-065

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options.

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 3 years.

% Historic Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options.

* Historical volatility is taken from old otr bond

Derivatives Strategy

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	Eurodollar													
	Jun 23	Jul 23	Aug 23	Sep 23	Dec 23	Mar 24	Jun 24	Sep 24	Dec 24	Mar 25	Jun 25	Sep 25	Dec 25	Mar 26
VOLATILITY														
Implied (%)														
Current	10.31	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (1d)	1.14	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day High	14.61	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day Low	9.11	-	-	-	-	-	-	-	-	-	-	-	-	-
Business Day	10.71	-	-	-	-	-	-	-	-	-	-	-	-	-
Historical (%)														
10 day	9.54	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	8.38	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	12.79	-	-	-	-	-	-	-	-	-	-	-	-	-
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)														
3 Year High	152.84	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	39.45	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	2.16	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	85.11	-	-	-	-	-	-	-	-	-	-	-	-	-
BASIS POINT VOLATILITY														
Implied (bp)														
Current	3.6	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (1d)	0.4	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (5d)	-0.4	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (20d)	-0.8	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day High	5.1	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day Low	3.2	-	-	-	-	-	-	-	-	-	-	-	-	-
Historical (bp)														
10 day	3.4	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	2.9	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	4.3	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (bp)														
3 Year High	28.4	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	2.4	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	0.2	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	22.3	-	-	-	-	-	-	-	-	-	-	-	-	-
Yield Change Required to... (bp)														
Offset 1 C-day Decay	3.1	-	-	-	-	-	-	-	-	-	-	-	-	-
Offset 1 B-day Decay	3.7	-	-	-	-	-	-	-	-	-	-	-	-	-
Offset 1 Month Decay	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Break Even at Expiry	8.0	-	-	-	-	-	-	-	-	-	-	-	-	-
BPV per Contract	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00
Cal/Bus Days to Expiry	11/7	36/24	64/44	102/69	193/132	284/193	375/256	466/319	557/381	648/442	739/505	830/568	921/630	1012/691
Futures Price	94.460	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options.

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 3 years.

% Historic Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options.

Derivatives Strategy

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	Eurodollar 1Yr MidCurve					Eurodollar 2Yr MidCurve				Eurodollar 3Yr MidCurve				Eurodollar 4Yr MidCurve				Eurodollar 5Yr MidCurve			
	Jun 23	Jul 23	Aug 23	Sep 23	Dec 23	Jun 23	Jul 23	Aug 23	Sep 23	Jun 23	Jul 23	Aug 23	Sep 23	Jun 23	Jul 23	Aug 23	Sep 23	Jun 23	Jul 23	Aug 23	Sep 23
VOLATILITY																					
Implied (%)																					
Current	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (1d)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Business Day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Historical (%)																					
10 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)																					
3 Year High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BASIS POINT VOLATILITY																					
Implied (bp)																					
Current	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (1d)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (5d)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Change (20d)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Historical (bp)																					
10 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
50 day	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (bp)																					
3 Year High	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Average	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3 Year Low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
% Historic Above Implied	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Yield Change Required to... (bp)																					
Offset 1 C-day Decay	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Offset 1 B-day Decay	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Offset 1 Month Decay	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Break Even at Expiry	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BPV per Contract	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00
Cal/Bus Days to Expiry	8/6	36/24	64/44	99/68	190/131	8/6	36/24	64/44	99/68	8/6	36/24	64/44	99/68	8/6	36/24	64/44	99/68	8/6	36/24	64/44	99/68
Futures Price	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options.

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 3 years.

% Historic Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options.

Derivatives Strategy

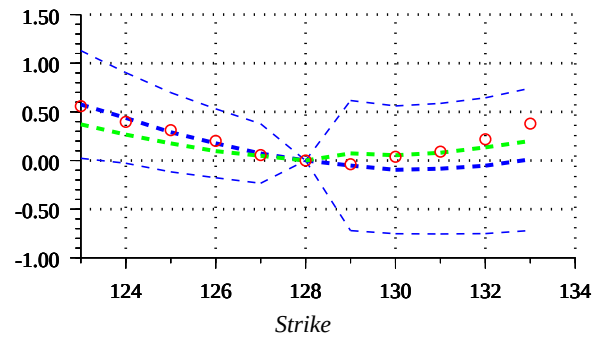
jpmorganmarkets.com

September 2023 Treasury Bond

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
123 P	0.56	0.58	0.37	11.7	-0.02	0.19	-0.06
124 P	0.40	0.44	0.27	12.8	-0.04	0.13	-0.17
125 P	0.31	0.29	0.18	13.7	0.02	0.14	0.10
126 P	0.20	0.18	0.10	14.4	0.02	0.11	0.14
127 P	0.06	0.07	0.05	14.8	-0.02	0.01	-0.10
129 C	-0.04	-0.05	0.07	14.7	0.01	-0.11	0.04
130 C	0.04	-0.10	0.05	14.1	0.13	-0.02	0.41
131 C	0.09	-0.08	0.08	13.4	0.18	0.01	0.52
132 C	0.22	-0.05	0.14	12.4	0.27	0.08	0.78
133 C	0.38	0.01	0.20	11.4	0.37	0.18	1.00

Sep 23 Bond futures @ 127-20; ATM imp. vol. = 11.41%; Vol. beta = -0.2515;

Skew (price vol above ATM)

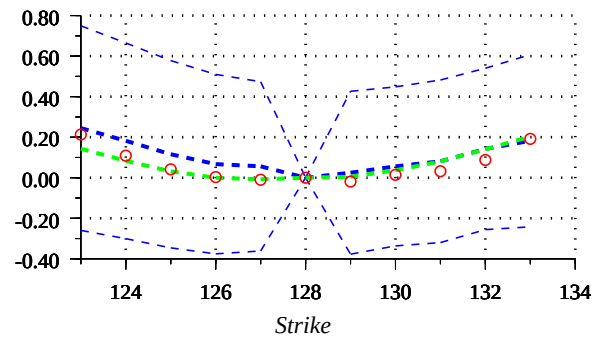


December 2023 Treasury Bond

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
123 P	0.21	0.24	0.14	19.0	-0.03	0.07	-0.13
124 P	0.11	0.18	0.08	19.9	-0.07	0.03	-0.30
125 P	0.04	0.12	0.03	20.6	-0.07	0.01	-0.32
126 P	0.00	0.07	-0.00	21.2	-0.06	0.00	-0.29
127 P	-0.01	0.06	-0.01	21.5	-0.07	-0.00	-0.32
129 C	-0.02	0.02	0.00	21.5	-0.04	-0.02	-0.22
130 C	0.01	0.06	0.04	21.2	-0.04	-0.02	-0.21
131 C	0.03	0.08	0.08	20.7	-0.05	-0.05	-0.24
132 C	0.09	0.14	0.14	20.1	-0.05	-0.05	-0.27
133 C	0.19	0.18	0.20	19.3	0.01	-0.01	0.04

Dec 23 Bond futures @ 127-21; ATM imp. vol. = 11.49%; Vol. beta = -0.1015;

Skew (price vol above ATM)

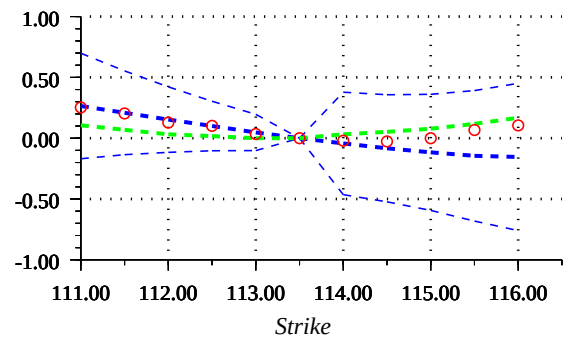


September 2023 Treasury Note

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
111.00 P	0.25	0.26	0.11	10.3	-0.01	0.15	-0.05
111.50 P	0.20	0.21	0.07	11.2	-0.01	0.13	-0.03
112.00 P	0.13	0.15	0.03	11.9	-0.03	0.10	-0.19
112.50 P	0.10	0.10	0.02	12.5	0.00	0.08	0.01
113.00 P	0.03	0.05	-0.00	13.0	-0.01	0.03	-0.17
114.00 C	-0.02	-0.04	0.03	13.2	0.02	-0.05	0.11
114.50 C	-0.03	-0.08	0.05	13.0	0.06	-0.08	0.26
115.00 C	-0.00	-0.12	0.08	12.6	0.12	-0.08	0.49
115.50 C	0.07	-0.14	0.12	12.1	0.21	-0.05	0.79
116.00 C	0.11	-0.15	0.17	11.4	0.26	-0.06	0.86

Sep 23 Note futures @ 113-235; ATM imp. vol. = 7.33%; Vol. beta = -0.1637;

Skew (price vol above ATM)

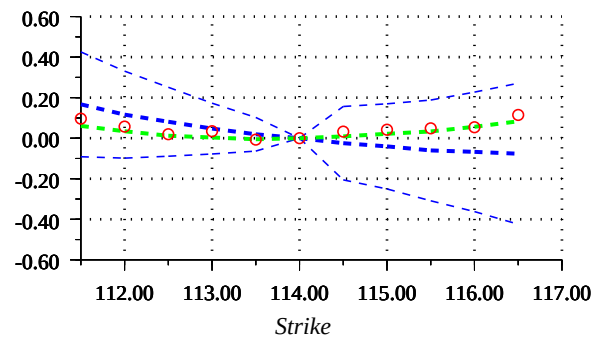


December 2023 Treasury Note

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
111.50 P	0.10	0.17	0.06	17.1	-0.07	0.04	-0.55
112.00 P	0.06	0.12	0.03	17.8	-0.06	0.02	-0.55
112.50 P	0.02	0.08	0.01	18.3	-0.06	0.01	-0.72
113.00 P	0.03	0.05	0.00	18.8	-0.01	0.03	-0.21
113.50 P	-0.01	0.02	-0.01	19.1	-0.03	-0.00	-0.64
114.50 C	0.03	-0.02	0.01	19.3	0.06	0.02	0.63
115.00 C	0.04	-0.04	0.02	19.2	0.08	0.02	0.78
115.50 C	0.05	-0.06	0.03	19.0	0.11	0.02	0.88
116.00 C	0.05	-0.07	0.06	18.6	0.12	-0.00	0.82
116.50 C	0.11	-0.08	0.08	18.1	0.19	0.03	1.10

Dec 23 Note futures @ 114-075; ATM imp. vol. = 7.43%; Vol. beta = -0.1698;

Skew (price vol above ATM)



- Our model takes into account the following factors over the last 3 years to explain the skew:
 - The number of business days to option expiration.
 - The level of the J.P. Morgan Emerging Markets Bond Index (EMBI).
 - The price of the underlying future.
 - The 'volatility beta', which represents how much at-the-money implied volatility changes with changes in the level of the underlying futures. It is computed using the last 20 business days of data.
 - Volatility of implied volatility over the last 60 business days.
- The average skew is computed using the last 125 business days of data.
- The model rolls to the next contract 5 business days before option expiration.
- Vega is the change in the option price (in ticks) due to a 1% change in implied volatility.

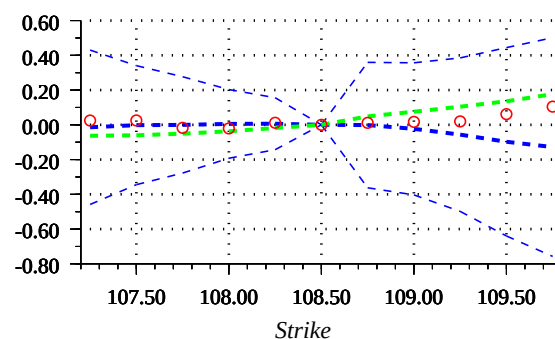
○ Actual
--- Average
--- Model
--- Model +/- 2s.d.

September 2023 5 Year Treasury Note

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
107.25 P	0.02	-0.01	-0.06	11.4	0.04	0.09	0.17
107.50 P	0.03	-0.00	-0.06	11.8	0.03	0.09	0.16
107.75 P	-0.02	-0.00	-0.05	12.1	-0.02	0.03	-0.12
108.00 P	-0.02	0.01	-0.04	12.4	-0.03	0.02	-0.26
108.25 P	0.01	0.01	-0.02	12.6	0.01	0.03	0.07
108.75 C	0.01	-0.00	0.05	12.6	0.01	-0.04	0.07
109.00 C	0.02	-0.02	0.08	12.5	0.04	-0.06	0.21
109.25 C	0.02	-0.06	0.10	12.3	0.08	-0.08	0.34
109.50 C	0.06	-0.10	0.14	12.0	0.16	-0.07	0.59
109.75 C	0.10	-0.13	0.18	11.7	0.23	-0.07	0.74

Sep 23 5-Year futures @ 108-17; ATM imp. vol. = 5.69%; Vol. beta = -0.0190;

Skew (price vol above ATM)

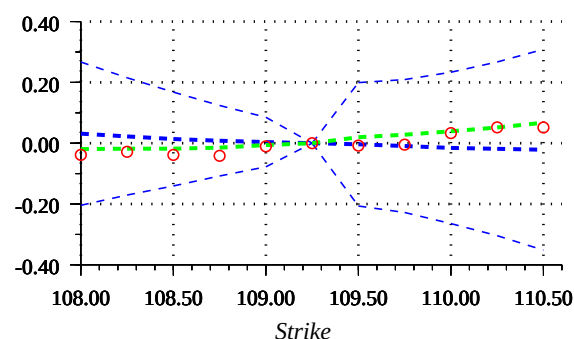


December 2023 5 Year Treasury Note

Strike	Actual	Model	Avg.	Vega	Act.-Mod.	Act.-Avg.	Z-Score
108.00 P	-0.04	0.03	-0.02	17.6	-0.07	-0.02	-0.59
108.25 P	-0.03	0.02	-0.02	17.9	-0.05	-0.01	-0.52
108.50 P	-0.04	0.01	-0.02	18.2	-0.05	-0.02	-0.68
108.75 P	-0.04	0.01	-0.01	18.3	-0.05	-0.03	-0.85
109.00 P	-0.01	0.00	-0.01	18.4	-0.01	-0.00	-0.35
109.50 C	-0.01	-0.00	0.02	18.5	-0.00	-0.03	-0.04
109.75 C	-0.00	-0.01	0.03	18.4	0.00	-0.03	0.04
110.00 C	0.03	-0.02	0.04	18.2	0.05	-0.01	0.39
110.25 C	0.05	-0.02	0.05	18.0	0.07	0.00	0.49
110.50 C	0.05	-0.02	0.07	17.8	0.07	-0.02	0.44

Dec 23 5-Year futures @ 109-065; ATM imp. vol. = 5.65%; Vol. beta = -0.1169;

Skew (price vol above ATM)



1. Our model takes into account the following factors over the last 3 years to explain the skew:

- The number of business days to option expiration.
- The level of the J.P. Morgan Emerging Markets Bond Index (EMBI).
- The price of the underlying future.
- The 'volatility beta', which represents how much at-the-money implied volatility changes with changes in the level of the underlying futures. It is computed using the last 20 business days of data.
- Volatility of implied volatility over the last 60 business days.

2. The average skew is computed using the last 125 business days of data.

3. The model rolls to the next contract 5 business days before option expiration.

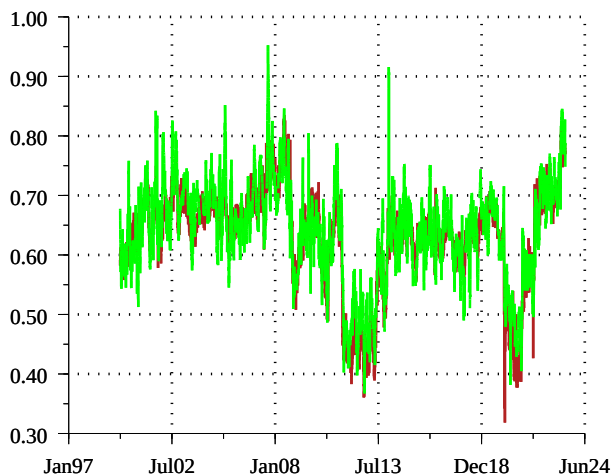
4. Vega is the change in the option price (in ticks) due to a 1% change in implied volatility.

○ Actual
--- Model
--- Average
--- Model +/- 2s.d.

Maturity Structure of Treasury Volatility

Ratio of Price Volatility

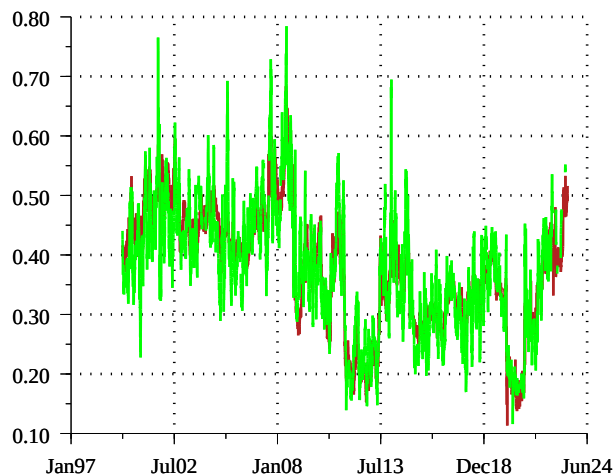
(Front 5yr)/(Front 10yr)



	Current	1 Yr Avg
Implied	0.78	0.73
Historical	0.75	0.74

Ratio of Price Volatility

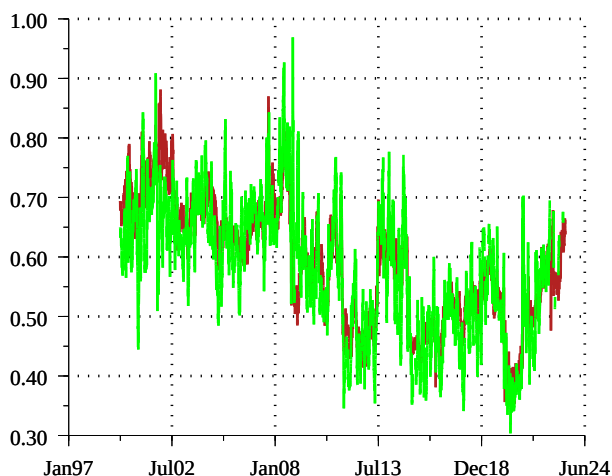
(Front 5yr)/(Front 30yr)



	Current	1 Yr Avg
Implied	0.50	0.43
Historical	N/A	0.45

Ratio of Price Volatility

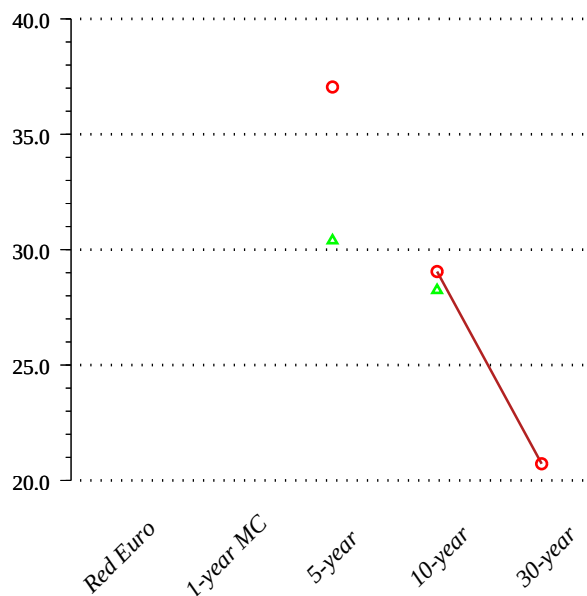
(Front 10yr)/(Front 30yr)



	Current	1 Yr Avg
Implied	0.64	0.59
Historical	N/A	0.62

Maturity Structure of Yield Volatility

Volatility



—○— Implied
—▲— 20-day Historical

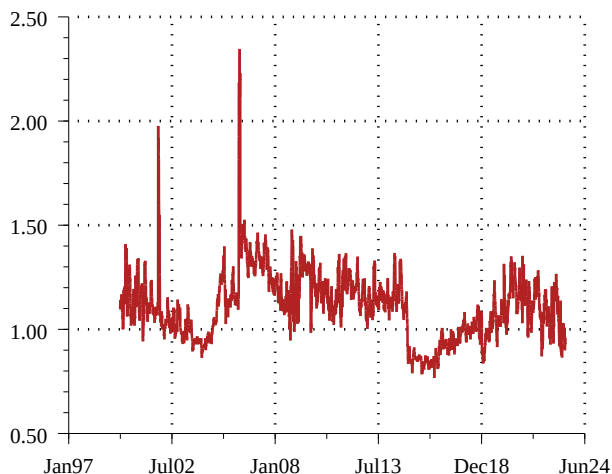
— Implied
— 20-day Historical

1.Contract switches 20 business days before quarterly option expiration.

Treasury OTC and Exchange Volatility Report

OTC 30's / CBOT 30's

Ratio Price Volatility



	Current	6M Avg	FV
Implied	N/A	N/A	0.84
Historical	0.93	1.02	

OTC 10's / CBOT 10's

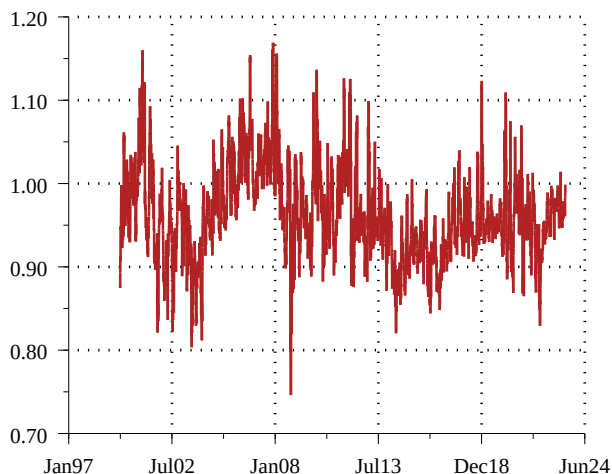
Ratio Price Volatility



	Current	6M Avg	FV
Implied	N/A	N/A	1.01
Historical	1.04	1.06	

OTC 5's / CBOT 5's

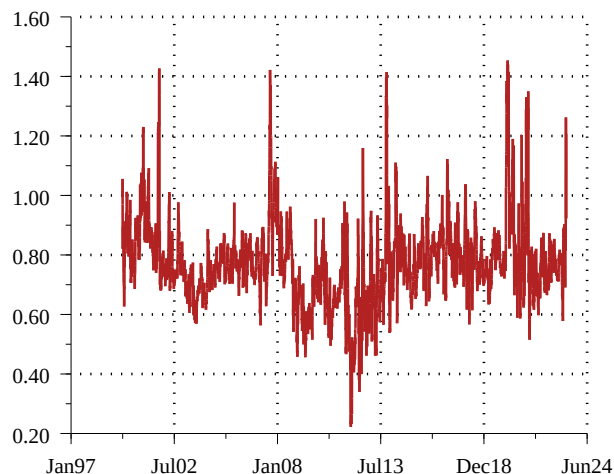
Ratio Price Volatility



	Current	6M Avg	FV
Implied	N/A	N/A	0.96
Historical	1.00	0.97	

OTC 2's / 1-Yr Eurodollar midcurves

Ratio Price Volatility



	Current	6M Avg	FV
Implied	N/A	N/A	N/A
Historical	N/A	0.80	

— Ratio Implieds
— Ratio 20-day Historicals

- Contract switches 30 business days before quarterly option expiration.